

Aman Singh

Junior Network Engineering & Automation

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OBJECTIVE

Specialising in network automation and data center architecture. I build spine-leaf fabrics, SDN controllers, and automated lab environments from scratch. Goal: work at the intersection of infrastructure and intelligence, designing networks that are faster, smarter, and more resilient. Actively pursuing CCNA. Looking for a team where I can grow fast and help build something that matters.

TECHNICAL SKILLS

Protocols	BGP (eBGP/iBGP), OSPF, VXLAN, EVPN, OpenFlow, ECMP, STP
Automation	Ansible, Containerlab, FRRouting (FRR), Infrastructure as Code (YAML)
SDN / Virt	Mininet, Open vSwitch, Faucet Controller, Docker, Docker Compose
Observability	Prometheus, Grafana, cAdvisor
Programming	Python (network scripting), Bash / Shell
OS / Tools	Kali Linux, Ubuntu, Git, GitHub, VS Code

PROJECTS

Network Lab Orchestrator

github.com/MrWater00/network-lab-orchestrator

Python · Ansible · Containerlab · Prometheus · Grafana · Docker

- Automated IaC platform that deploys, configures & monitors full network topologies (routers, switches, hosts) from a single YAML file using Ansible + FRRouting.
- Integrated Prometheus + Grafana + cAdvisor observability stack for real-time CPU, memory & network I/O across all lab containers.

Spine-Leaf Data Center Fabric

github.com/MrWater00/spine-leaf-fabric

Shell · Containerlab · FRRouting · eBGP · VXLAN · BGPEVPN

- Production-grade 2-spine / 2-leaf Clos fabric (same architecture as AWS & Google Cloud) with eBGP underlay, ECMP load balancing, and VXLAN/BGP EVPN overlay.
- Eliminated MAC flooding and achieved clean L2 extension across an L3 fabric using BGP EVPN control plane.

SDN Load Balancer

github.com/MrWater00/sdn-load-balancer

Python · Mininet · OpenvSwitch · Faucet · OpenFlow1.3

- Round-robin load balancer distributing HTTP traffic from a single VIP across 3 backends using programmatic OpenFlow rules with IP/MAC rewrite at switch level.
- Independently debugged Python 3.13 compatibility issues, migrating from Ryu to Faucet and resolving all flow rule conflicts and ARP challenges.

CERTIFICATIONS & EDUCATION

Cisco CCNA Self-studying · In Progress

Covering routing & switching, BGP, OSPF, VLANs, STP, ACLs, NAT reinforced with live Containerlab labs.

BCA (Data Science + AI)

Completed May 2026 · Result Awaited

ADDITIONAL WORK

dns-labs, bgp-labs, shadowScan (Go), gopherShell (Go), HoneyPot (Python), ai-chatbot & more → github.com/MrWater00